

Engagements · Capacity

BreakPoint Capacity: process capability assessment.

We don't ask only whether a plant can reach a flow. We determine the operating envelope within which it can reliably meet its treatment obligations, which constraint governs each edge, and how well each edge is known.

Five numbers that get confused

Ask a plant what its capacity is and you get a number. Ask five people at the utility and you get five, each correct, because "capacity" is doing the work of five distinct concepts. Most capacity studies report one and never say which. The assessment reports all five; the gaps between them are where the money is.

Nominal	Design flow: the figure on the drawings and in the licence. A statement of intent.	Literature-supported or Assumed
Theoretical	What unit calculations say each stage could handle under stated conditions, before coupled constraints.	Calculated
Demonstrated	What the plant has actually achieved and sustained while meeting its obligations, from the operating record.	Measured
Constrained	The flow at which the first real-world limitation binds: solids, chemistry, hydraulics, equipment, control, people.	Calculated or Modelled, calibrated
Resilient	What can be sustained with a unit out, at tail water quality, or under a combination of adverse conditions.	Modelled, scenarios stated

Two lenses, one envelope

Top-down: engineering analysis

Design basis, criteria, drawings, duties and control philosophy. Hydraulics, unit loadings, contact time, mass balance, kinetics, N-1, interfaces. Finds theoretical limits, missing redundancy and interactions. Alone, it describes a plant that has never been operated.

Bottom-up: operating evidence

SCADA, laboratory, operator logs, maintenance and production records. Percentile and event analysis, empirical envelopes, calibrated models. Finds actual behaviour, hidden setpoints, batch cycles and evidence gaps. Alone, it describes only conditions the plant has already seen.

Every material claim carries a BreakPoint evidence class, and a claim cannot move up the list without new evidence. We start with the operating record, not the design.

M

 Measured

C

 Calculated

L

 Literature-supported

Mo

 Modelled

I

 Inferred

A

 Assumed

H

 Heuristic

Capacity is an envelope, not a number.

Six stages. The same five steps.

Frame, Evidence, Reason, Challenge, Decide, applied to the capacity question. Staged so early findings improve the next round and the final models stay traceable to plant evidence.

01 · Frame

Frame the decision and the service outcome

Horizon, required output, obligations, operating philosophy, reliability target, outages, the decisions to support. Which of the five capacities does the decision turn on?

Output: scope, decision questions, success criteria, boundaries.

02 · Evidence

Reconstruct the plant and the evidence model

Map the train, units, recycle paths, instruments and lab points. Governed data model with tag mapping, quality flags, time alignment, lineage. Check the instruments before trusting the data.

Output: schematic, evidence register, data dictionary, quality audit.

03 · Evidence, Reason

Establish current operating capability

Flows, loads, water quality, production, operating states, performance. Segment events. This is where demonstrated capacity is found.

Output: baseline, event library, operating envelopes, data gaps.

04 · Reason

Model processes, balances and interfaces

Calculations and plant-specific models. Close water and solids balances; test loading, kinetics, contact time, headloss, residuals, recycle; find the coupled constraints. Reconcile with operators.

Output: unit checks, calibrated models, constraint map, confidence statement.

05 · Challenge

Stress test future and upset scenarios

Forecasts with seasonal conditions, p95 and p99 quality, outages, N-1, blends, settings, options. Which assumption is load-bearing? Where is absence of evidence posing as adequacy?

Output: scenario matrix, capability trajectory, thresholds, sensitivities.

06 · Decide

Prioritise actions and make it reusable

Separate operational changes from investigations and capital. Trials, stop criteria, instrumentation. Deploy as a refreshable report, dashboard or tool.

Output: roadmap, test plan, business case evidence, reusable product.

Three ways to engage

Capacity Screen

One plant or unit · 2 to 4 weeks · fixed price

The five capacities, the binding constraint at each edge, the evidence class of each, the gaps.

- Targeted evidence pack
- Constraint map
- Gaps ranked by decision relevance

Capacity Assessment

One plant · detailed

Calibrated models, closed balances, future and upset scenarios, prioritised roadmap.

- Calibrated models, thresholds
- Scenario matrix, trajectory
- Trial plans, business case evidence

Capacity Programme

Portfolio · recurring · hosted or licensed

The same logic as a refreshable dashboard or tool across a plant or portfolio.

- Governed evidence model
- Automated reports
- Capacity module in WaterPoint and AquaPoint

What the approach has done

Portfolio-scale capability. Automated assessment workflow and dashboard across a major utility's wastewater plants, first 27, now extending to every plant it operates.

Filtration capability from the record. A calibrated filtration model from thousands of filter runs, separating operational screening from guaranteed capacity.

Deferral through demonstrated headroom. Challenging a business-as-usual upgrade and identifying capital deferral by maximising existing assets.

Delivered by our principals in previous roles; not represented as BreakPoint deliveries.

Bring us the difficult question.